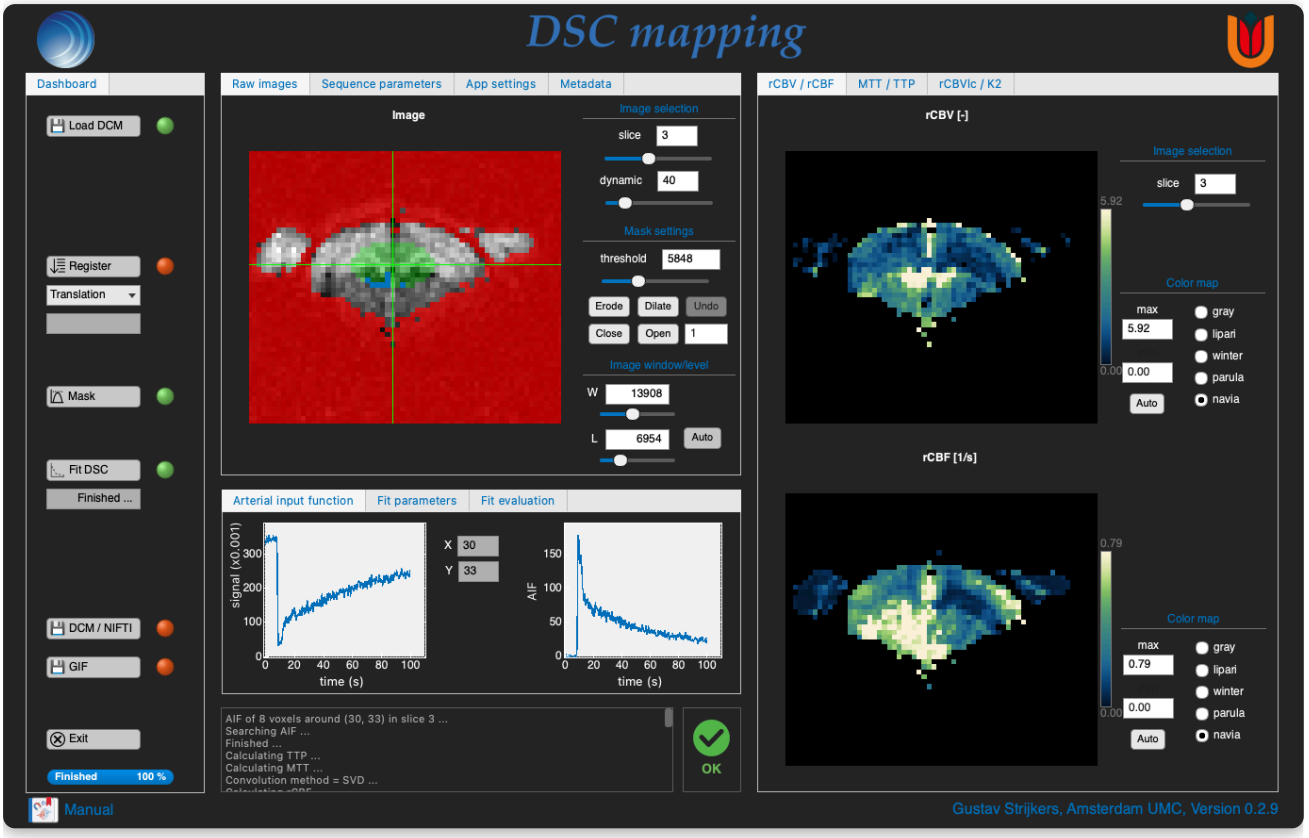


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1. Overview

DSCmapp is an application with a graphical user interface to map hemodynamic and vascular properties of brain tissue from a dynamic susceptibility contrast (DSC) MRI acquisition on an MR Solutions preclinical MRI system. It reads a DSC series of DICOM files, registers and masks the images, finds an arterial input function (AIF) around a point you click, and fits relative cerebral blood volume and flow, mean transit time, time to peak, leakage-corrected blood volume and the leakage constant K2. The maps can be exported as DICOM, NIfTI and GIF, each export with a metadata record and a report of how the maps were made.

DSCmapp is a stand-alone application: everything it uses, elastix and dcm2niix included, is inside it, and nothing else has to be installed.

2. Contact

DSCmapp and this manual are written by:

Gustav Strijkers
Amsterdam UMC
Biomedical Engineering and Physics
Amsterdam, the Netherlands
g.j.strijkers@amsterdamumc.nl

3. Disclaimer

This is a beta version. The software has been tested using mouse and rat data acquired with an MR Solutions preclinical 7.0T/24cm system. However, this does not warrant that the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free. In case of questions or issues, please contact Gustav Strijkers.

4. Installation

macOS

Requirements: a Mac with Apple silicon (M1 or later), macOS 26 or later.

DSCmapp comes as a disk image, `DSCmapp-0.3.4-macOS-arm64.dmg`. Nothing else has to be installed: Python, elastix, dcm2niix and everything else the app uses are inside it.

1. Open the disk image and drag **DSCmapp** onto the Applications folder in its window.
2. Open it from Applications. The app is not signed with an Apple Developer ID, so the first time macOS refuses it ("Apple could not verify ..."); click **Done**. Then open **System Settings > Privacy & Security**, scroll down to the line saying DSCmapp was blocked, click **Open Anyway** and confirm. From then on it opens as any other app.

   Instead of step 2, the same in Terminal, once:

   ```
   xattr -dr com.apple.quarantine /Applications/DSCmapp.app
   ```

The first start can take a little longer: macOS checks the app once.

Windows

Requirements: a PC with Windows 10 or 11, 64-bit.

DSCmapp comes as an installer, `DSCmapp-0.3.4-setup.exe`. Nothing else has to be installed: Python, elastix, dcm2niix and everything else the app uses are inside it.

1. Run the installer. It is not signed, so the first time Windows may stop it ("Windows protected your PC"): click **More info** and then **Run anyway**.
2. The installer puts the app in `%LOCALAPPDATA%\Programs\DSCmapp` for the current user, without administrator rights (installing it for all users is an option), with a Start-menu entry and, if you choose, a desktop shortcut. At the end it can start the app.

To remove the app, use **Uninstall DSCmapp** in the Start menu, or **Settings > Apps**.

5. Running the software

Open **DSCmapp** from the Applications folder (macOS), or from the Start menu or the desktop shortcut (Windows). The DSCmapp logo appears while the app loads, and the window follows within seconds. At its start the app looks for elastix, and the message box reports what it found; "Ready" says it is done. Only **Load DCM** and **Exit** are on until a dataset is loaded.

A series given on the command line (the folder of its `.dcm` files or one of them, as a program that starts DSCmapp with a series gives it) is loaded at once, as with **Load DCM**; a path the app cannot load gives a warning in the message box. When the app is open already, a scan given this way goes to the open window instead: the window comes to the front and loads it, or loads it when the step under way ends.

Newer versions. At its start the app asks GitHub (https://github.com/Moby1971/MRI-apps) whether there is a newer release of it. The message box then shows the current and the newest version and, when there is a newer one, where to download it. Without a network connection it says that the version could not be checked, and the app works as usual. `checkForUpdates` in the settings file switches the check off; `versionCheckTimeout` sets how many seconds it waits for GitHub (5).

Updates (App settings tab):

- **auto update**, ticked as it comes: new versions of DSCmapp arrive by themselves. When one is ready, a window says so with what is new: **Install and Restart** installs it and starts the new DSCmapp; **Later** installs it when you quit DSCmapp, without asking again. The window waits while DSCmapp is working. Without an internet connection nothing is said and nothing waits. On Windows, DSCmapp installed for all users can be updated only by an administrator; for other users the message box says once where to get the new version to install by hand.
- **Check now** looks for a newer version at once, also with **auto update** off; the message box says what it found. On the Mac, **Check for Updates...** in the DSCmapp menu does the same.
- **notify at start**, ticked as it comes, counts only with **auto update** off: when a newer version is available, a window at the start offers it. **Download** gets the installer for this computer, **Skip this version** shows no window for that version again, **Later** asks again at the next start.

The message box says at every start whether DSCmapp is up to date.

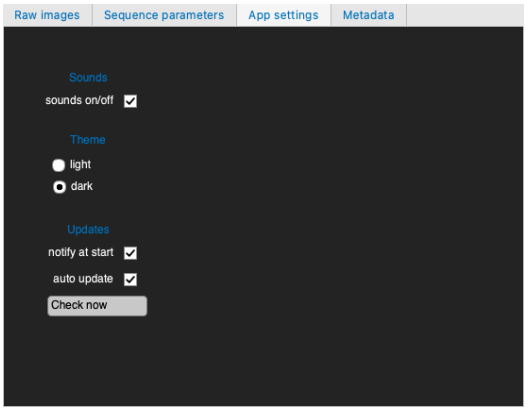
This manual. The **Manual** button at the bottom of the window, with the app's icon on a book, opens this manual in the computer's PDF viewer.

The window has a fixed size. On a screen too small for it, the window fits the screen and can be scrolled.

Settings and log files

DSCmapp keeps its files in a folder of its own, which it makes at its first start:

macOS	`~/Library/Application Support/DSCmapp/0.3.4p`
Windows	`%APPDATA%\DSCmapp\0.3.4p`



Settings. The settings are kept in `dscmappSettings.json`. The file is written with the standard values the first time a version starts, and can then be edited by hand; changes take effect at the next start. It holds the version check, the folders where loading and the exports start (`dicomImportPath`, `defaultExportPath`), the theme, the sounds, the registration method, the fit parameters and the color maps of the six maps, and the number of log files kept. A value that cannot be used is replaced by its standard value, and the message box says so; a file that cannot be read at all is set aside as `dscmappSettings-unreadable-<time>.json` and replaced by a new one. The App settings tab switches the theme, the sounds, notify at start and auto update.

DSCmapp keeps your choices of the theme, the sounds, notify at start, auto update, the color maps, the registration method, the SVD type, **kh** and **rho**, **quadratic R2* [CA]**, the **first dynamic**, the size of the AIF search area and the maximum number of AIF voxels, and the folders last used for loading and for exports, for the next start; a new version takes them over.

Log files. Every session writes a log file, `dscmapp-log-<date and time>.txt`, to the same folder. The log keeps every message and the parameters of each registration and fit. Logs older than the newest 50 are removed at start.

6. Basic operation



The DSCmapp program operates from a single window with 5 panels.

Panel 1: Dashboard. The task buttons that control the process, from top to bottom; registration can be left out. A green light shows a task done, yellow a task running, and red one not done or failed. At the bottom of the Dashboard one progress bar shows the progress of every task ("Loading 40 %", "Finished").

Panel 2: Raw images / Sequence parameters / App settings / Metadata. The images of the series, the relevant acquisition parameters, the theme and sounds of the app, and the record of the scan: the animal, the session and the contrast agent (step 7).

Panel 3: Arterial input function / Fit parameters / Fit evaluation. The AIF found, the fit's settings, and the fit of the voxel clicked.

Panel 4: rCBV / rCBF, MTT / TTP, rCBVlc / K2. The calculated parameter maps, in 3 tabs of 2 maps each.

Panel 5: Messages. The program messages, newest on top, and the status: OK, BUSY, WARNING or ERROR.

Step 1: Loading data

Press **Load DCM** to load the DICOM files: choose the folder that holds the series' `.dcm` files (…`/DICOM/<series>/1` in the scanner's export). On the scanner computer (Windows) the first dialog of a session opens in the newest exam of the scanner's Data folder, unless the settings name a folder (`dicomImportPath`). Elsewhere it opens in the folder loaded last.

The app expects multi- or single-slice brain data with at least 50 dynamics (time series), acquired during the injection of a contrast agent. The files are checked: every dynamic has to hold the slice positions of the first. A series that does not pass is refused, with the reason in the message box, and the dataset loaded before stays.

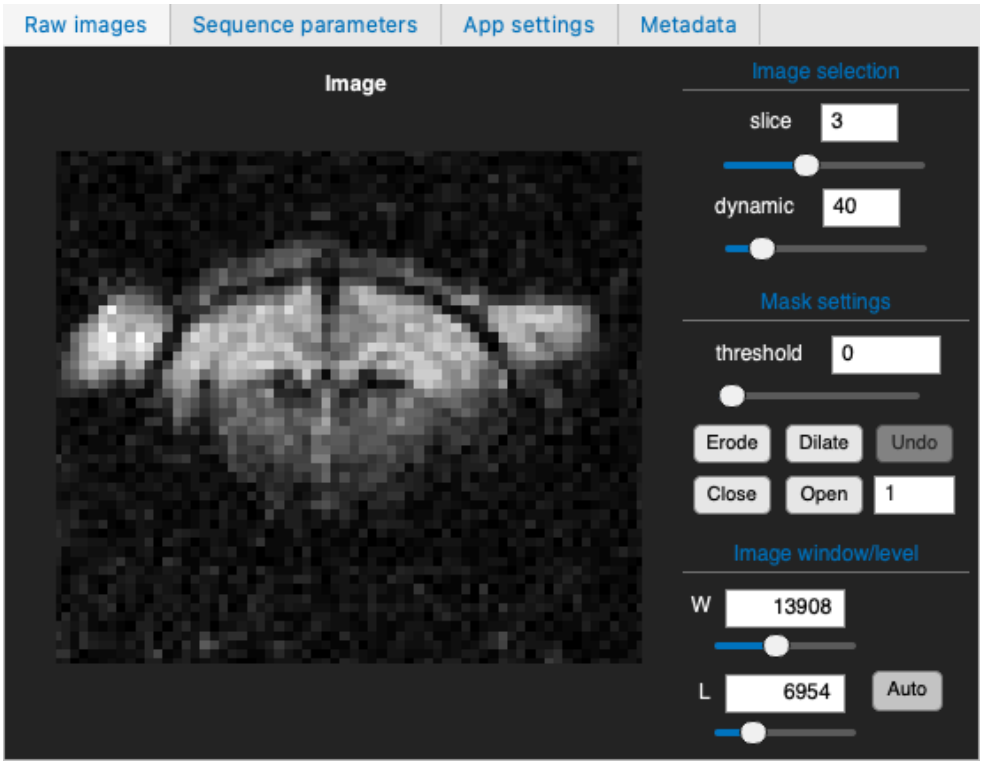
The relevant acquisition parameters are shown in the Sequence parameters tab.



By default the analysis starts at dynamic 15. If needed, this can be changed with **first dynamic** in the Fit parameters tab, which clears a registration, the mask, the AIF and the maps.

Step 2: Image registration

Use the sliders and fields of the Raw images tab to inspect the slices and dynamics. The slices are numbered as on the scanner. The image scale can be adjusted with the window (**W**) and level (**L**) fields and sliders, or with **Auto**.



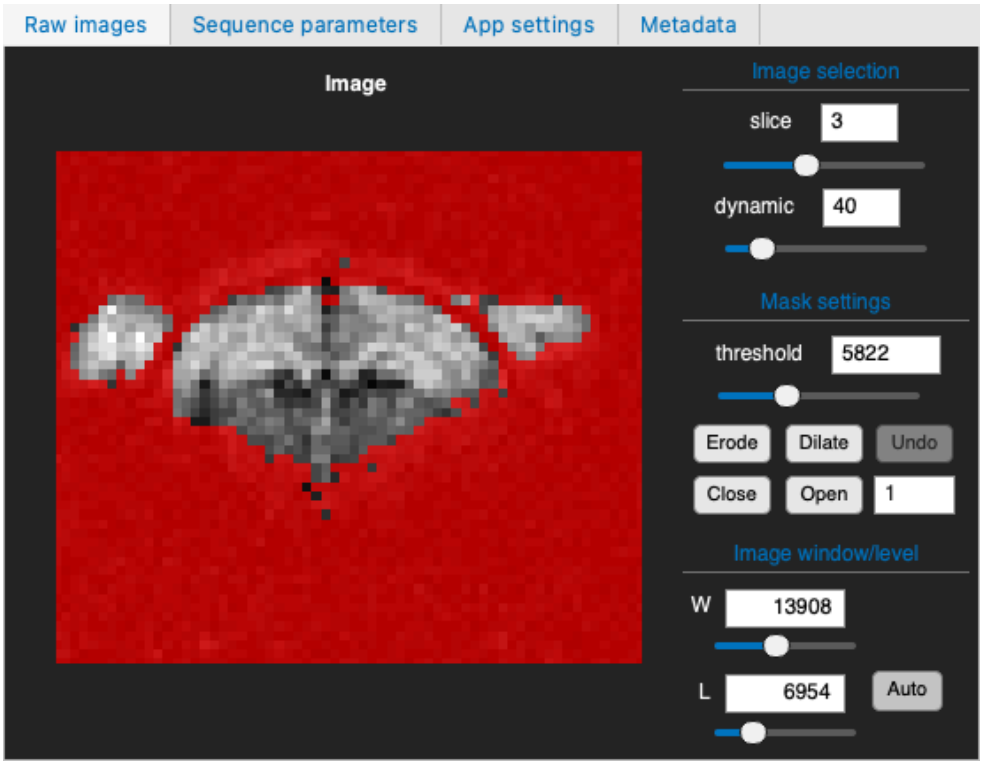
If the images are not perfectly registered, press **Register** to register them with elastix, as a **Translation**, **Rigid**, **Affine** or **B-Spline** transform (the list below the button). While the registration runs, the button reads **Stop**; pressing it stops the registration and leaves the images, the mask, the AIF and the maps as they were before you pressed **Register**.

NOTE: usually not needed, slow for large series, and it clears the mask, the AIF and the maps.

Step 3: Segmenting the images

To isolate the brain the images can be segmented. Press **Mask** for an automatic threshold (Otsu) per slice, on the first dynamic of the analysis. The red pixels indicate the regions that will be set to zero in the DSC parameter maps.

If the automatic threshold is not optimal, it can be adjusted per slice with the **threshold** field or slider. The segmentation applies to all dynamics and can be refined by **Erode**, **Dilate**, **Close** and **Open** operations on the red region, of the size in the field beside them; **Undo** takes the last one back.

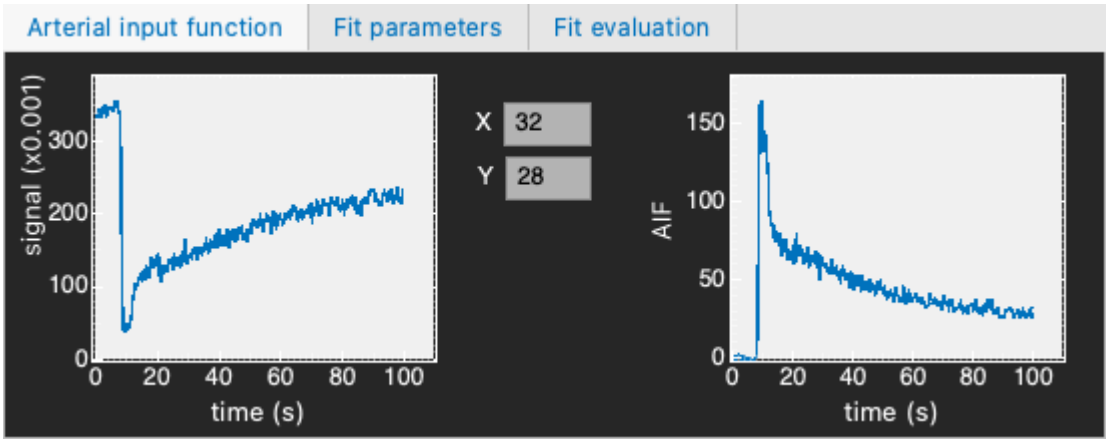


Step 4: Selecting an arterial input function (AIF)

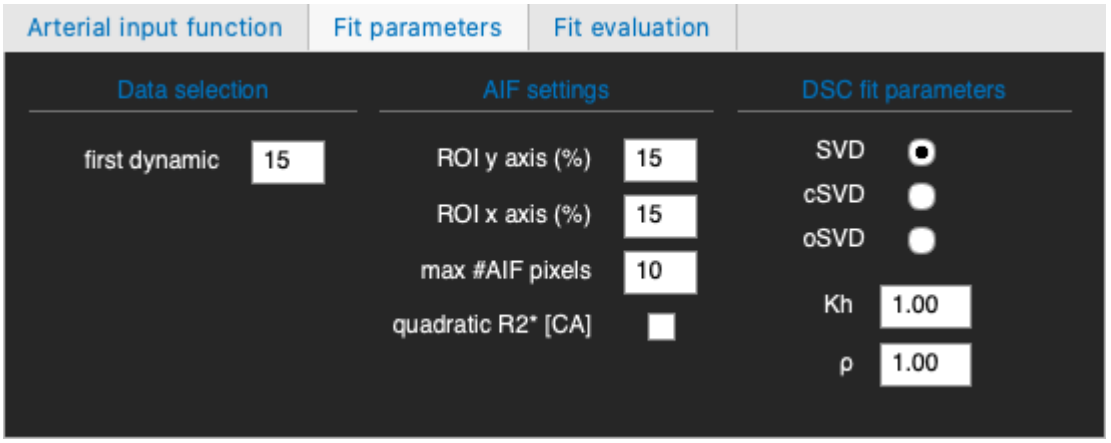
To semi-automatically determine an arterial input function, click on the brain in the image. The AIF is searched in the slice shown, around the voxel clicked.

To determine the correct AIF the following approach should be considered. Make a selection for which the signal change is early (first to respond), deep (substantial signal loss), and narrow. Selecting a voxel within an artery might not always be advisable because of signal dephasing by a high contrast agent concentration in the artery. It is acceptable to select voxels adjacent to the artery as long as the local signal drop is early, deep and narrow.

The AIF is selected automatically from the green-shaded area. The Arterial input function tab shows the mean signal of the AIF's voxels and its concentration; the voxels themselves are shown in blue in the image.



When no AIF is found around the voxel, the message box says why and to try another point, and the AIF found before is kept. The size of the green-shaded search area and the maximum number of AIF voxels can be adjusted in the Fit parameters tab.



Step 5: Fitting the hemodynamic parameters

The app estimates the CBF by solving the convolution equation

> $c(t) = CBF \cdot \int AIF(\tau)\, R(t - \tau)\, d\tau$

with c(t) the tissue contrast concentration, AIF(t) the arterial input function, and R(t) the fraction of contrast agent present in the volume of interest (the residue function).

The deconvolution uses a singular value decomposition (SVD), of one of three types chosen in the Fit parameters tab: standard SVD, circular SVD (cSVD) or oscillation-index SVD (oSVD).

For the AIF, a linear (not ticked) or quadratic (**quadratic R2* [CA]** ticked) dependency of R_2* on the contrast agent concentration can be chosen; it applies to every map. For quantitative perfusion values one can adjust ρ (**rho**), the density of brain tissue (typically 1 g/ml tissue), and Kh (**kh**), the hematocrit factor (typically 0.67 in the capillaries). kh/ρ scales rCBV, rCBVlc and rCBF alike; MTT, their ratio, does not change.

To start the fitting, press **Fit DSC**. The AIF is searched again at the point of the AIF shown. While the fit runs, the button reads **Stop**; pressing it stops the fit and leaves the maps and the parameters as they were. The application produces the following maps:

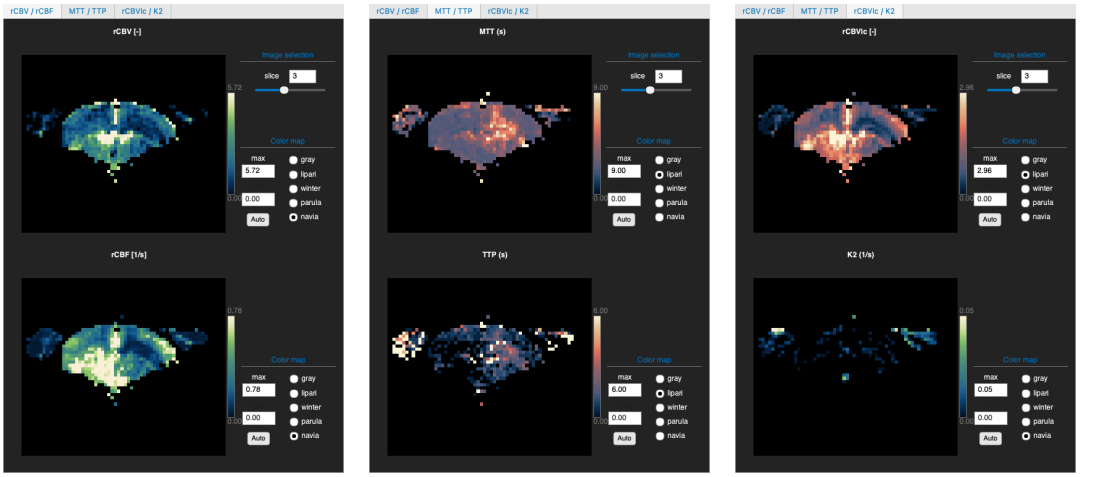
Map	
rCBV	relative cerebral blood volume [-]
rCBF	relative cerebral blood flow [1/s]
MTT	mean transit time [s], rCBVlc over rCBF
TTP	time to peak [s]: the delay between the peak of the AIF and the tissue's
rCBVlc	relative CBV with leakage correction [-] (Boxerman et al. 2006)
K2	leakage parameter [1/s]

A time to peak before the AIF's peak is set to 0.

Step 6: Inspecting the results

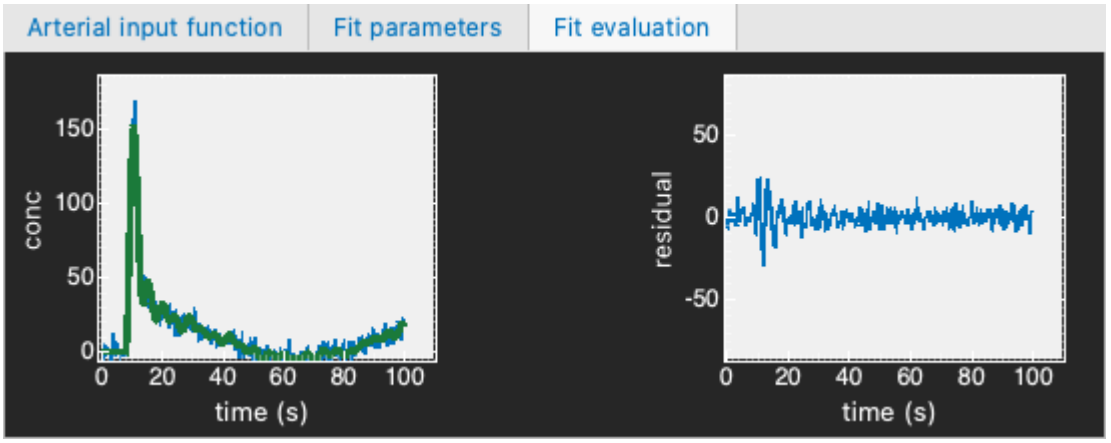
The hemodynamic parameter maps are shown in the panel on the right, two per tab. Each map has its own color map (gray, lipari, winter, parula or navia); the slice fields of the three tabs move together, and the mouse wheel over the maps steps through the slices. The mouse wheel over a slider moves that slider, one step per notch.

rCBV / rCBF **MTT / TTP** **rCBVlc / K2**



Each map has a **max** and a **min** for its color scale, which **Auto** sets to 3 times the median of the slice shown and to 0. The color bar stands beside each map, with its end values above and below it.

By clicking on the raw image after the fit, the Fit evaluation tab shows the concentration (blue) and the fit (green) of that voxel, and on the right their difference. The click also searches an AIF around the voxel, shown in the Arterial input function tab; the maps keep the AIF they were fitted with.

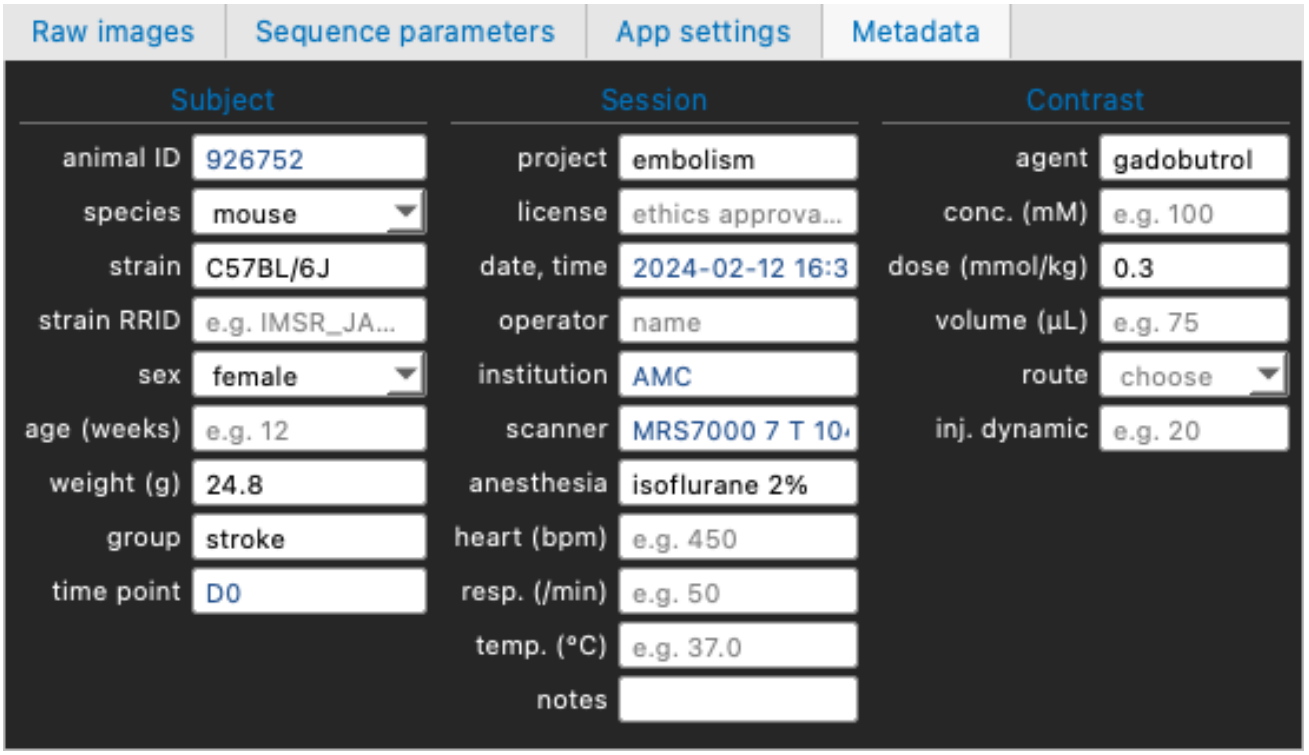


Step 7: Metadata

The **Metadata** tab, beside the Raw images, Sequence parameters and App settings tabs, holds the record of the scan, which every export writes with the maps. Its fields stand in three columns:

- **Subject:** animal ID, species, strain and its RRID, sex, age, weight, group and time point;
- **Session:** project, ethics license, date and time, operator, institution, scanner, anesthesia, heart and respiration rate, and temperature, with a field for notes;

- **Contrast:** agent, concentration, dose, volume, route and the dynamic it was injected at.



At a load the app fills in what the scanner's files say: the animal ID and the time point from the patient name typed at the console (`20240212_926752_D0`: animal 926752, day 0), the project, the date and time, the institution and the scanner. What the console leaves at its standard values (sex O, weight 0, the login as the operator) is left empty. A value the app filled in is shown in blue; every field can be corrected, and then counts as yours.

Your institution, operator, anesthesia, species, strain and strain RRID, and the contrast agent, its concentration and route, are kept for later exams: they are shown, in blue, from the start, and a scan's own value (from its files or its exam) takes their place where it has one.

What was filled in for an earlier scan of the same exam is filled in again (DSCmapp, Retrospective, P2ROUD and the other mapping apps alike): the animal's details are typed only once per exam. The date, the heart and respiration rate, the temperature, the notes and the injection's dynamic belong to one scan and are never taken from another.

Species, sex and route are lists to choose from, or to type in; numbers are kept with their units (a number field holding something else is edged in red). The species and sex are written with their codes (NCBI Taxonomy, PATO), so the record can be read by other software as well as by people.

Step 8: Exporting the maps

There are two ways to export the maps.

1. **DCM / NIFTI** — exports the six maps, masked, in DICOM and NIfTI format for further processing in third-party software. Each DICOM map is a series of its own, numbered after the scan's series (rCBV +1 … K2 +6). The DICOM files carry the species, strain, sex, age and weight of the animal and the contrast agent, where the Metadata tab has them.

2. **GIF** — exports the six maps as animated GIF files, one frame per slice, in the color maps and scales shown, with two pictures of the app's window.

When the DICOM files were loaded from the scanner's folders, the exports go beside them, in …/Data/<study>/DICOM, Nifti and GIF; otherwise the app asks for a folder, and starts at the folder chosen last. Every export gets a new numbered folder, so that no export overwrites another.

The export folders and files are named by the series number or the scan's name (<tag>); spaces and other special characters in it become _, so that the names have no spaces.

Every export also writes, in …/Data/<study>/Metadata/<tag>DSC/<n>:

- metadata.json — the Metadata tab's record;
- dscmapp-report_<tag>.json and .txt — the report of the export: the app and its libraries, the scan and a checksum of its files, its acquisition, every step of the session with its parameters (the registration, the mask and your changes to it, the AIF you clicked and the gamma variate fitted to it, the fit with all its settings), the statistics of each map per slice, and every file the export wrote with its checksum;
- ro-crate-metadata.json — the same as an RO-Crate (https://www.researchobject.org/ro-crate/), a standard description of the folder for data repositories.

The paths in them are relative to the study folder, so they hold no names of your own folders.

Step 9: Exit

Press **Exit** to shut down the program. Closing the window does the same, also while a step runs: a registration or a fit is stopped, the window disappears, and the program closes as soon as the step has ended.

7. Credits

- **DSC-MRI toolbox** — D. Peruzzo, A. Bertoldo, F. Zanderigo, C. Cobelli, Automatic selection of arterial input function on dynamic contrast-enhanced MR images, Comput Methods Programs Biomed 104(3), e148–157 (2011); M. Castellaro, D. Peruzzo et al., https://github.com/marcocastellaro/dsc-mri-toolbox.
- **SVD deconvolution** — L. Østergaard, R. M. Weisskoff, D. A. Chesler, C. Gyldensted, B. R. Rosen, High resolution measurement of cerebral blood flow using intravascular tracer bolus passages. Part I: Mathematical approach and statistical analysis, Magn Reson Med 36(5), 715–725 (1996); F. Zanderigo, A. Bertoldo, G. Pillonetto, C. Cobelli, Nonlinear stochastic regularization to characterize tissue residue function in bolus-tracking MRI: assessment and comparison with SVD, block-circulant SVD, and Tikhonov, IEEE Trans Biomed Eng 56(5), 1287–1297 (2009).
- **cSVD and oSVD** — O. Wu, L. Østergaard, R. M. Weisskoff, T. Benner, B. R. Rosen, A. G. Sorensen, Tracer arrival timing-insensitive technique for estimating flow in MR perfusion-weighted imaging using singular value decomposition with a block-circulant deconvolution matrix, Magn Reson Med 50(1), 164–174 (2003).
- **Leakage correction** — J. L. Boxerman, K. M. Schmainda, R. M. Weisskoff, Relative cerebral blood volume maps corrected for contrast agent extravasation significantly correlate with glioma tumor grade, AJNR Am J Neuroradiol 27(4), 859–867 (2006).
- **elastix** — S. Klein, M. Staring, K. Murphy, M.A. Viergever, J.P.W. Pluim, elastix: a toolbox for intensity based medical image registration, IEEE Transactions on Medical Imaging 29(1), 196–205 (2010). https://elastix.dev
- **Color maps** — F. Crameri, Scientific colour maps (lipari, navia), Zenodo, doi:10.5281/zenodo.1243862.
- **NIfTI conversion** — dcm2niix: C. Rorden et al., https://github.com/rordenlab/dcm2niix.